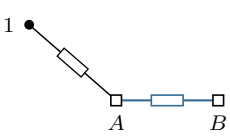
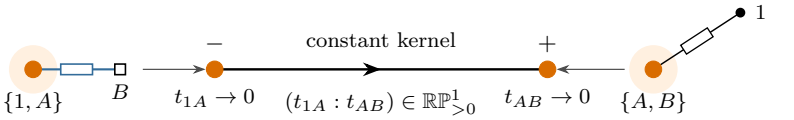
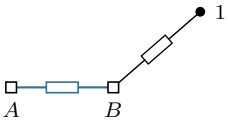
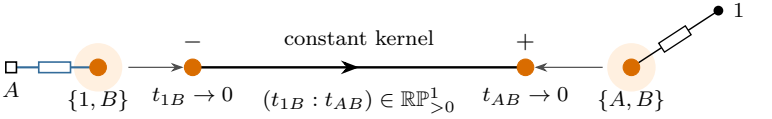
(a) One interaction node





$(Q_0A)B$ $\qquad$ $Q_0(AB)$, letter of A

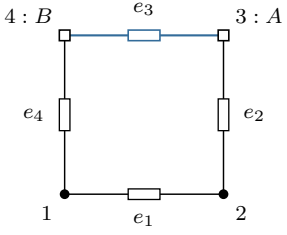




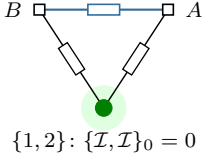
$A(Q_0B)$ $\qquad$ $Q_0(AB)$, letter of B

$$\sum_{\text{circuits}} \sum_{\text{ends}} = 0 \implies Q_0(AB) = (Q_0A)B + (-1)^{|A|}A(Q_0B)$$

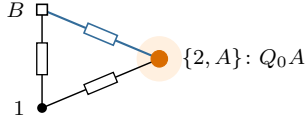
(b) Two interaction nodes: the marked square ring



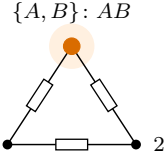
short e_1



short e_2

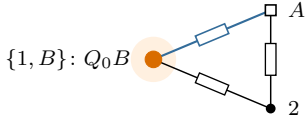


$(Q_0A) \star B$



short e_3

$Q_1(AB)$



short e_4

$A \star (Q_0B)$